

# Software Requirements Engineering (SRE)

## Assignment 2: Problem Analysis in a Government Organization (CLO-2)

### Group# 19

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### Step# 1: Problem Definition

**Problem definition:** Delay in arrival and departure of passenger trains.

#### Problem statement table:

| <u>Element</u>    | <u>Description</u>                                                                      |
|-------------------|-----------------------------------------------------------------------------------------|
| The problem of... | Delay in the arrival and departure of passenger trains.                                 |
| Affects...        | <ul style="list-style-type: none"><li>• Travelers.</li><li>• Railway workers.</li></ul> |

|                                  |                                                                                                                                                                                                                                                                              |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | <ul style="list-style-type: none"> <li>• Ministry of Finance of Pakistan.</li> <li>• Railway Management team.</li> </ul>                                                                                                                                                     |
| <b>And results in...</b>         | <ul style="list-style-type: none"> <li>• Excessive loss of time (both passengers and railway staff).</li> <li>• Dissatisfaction of the public passengers.</li> <li>• Loss in revenue of the government of Pakistan.</li> <li>• Overcrowding of public passengers.</li> </ul> |
| <b>Benefits of a solution...</b> | <ul style="list-style-type: none"> <li>• Gain of revenue for the government of Pakistan.</li> <li>• The least amount of crowd will be created.</li> <li>• Arrivals of passengers on proper time.</li> </ul>                                                                  |

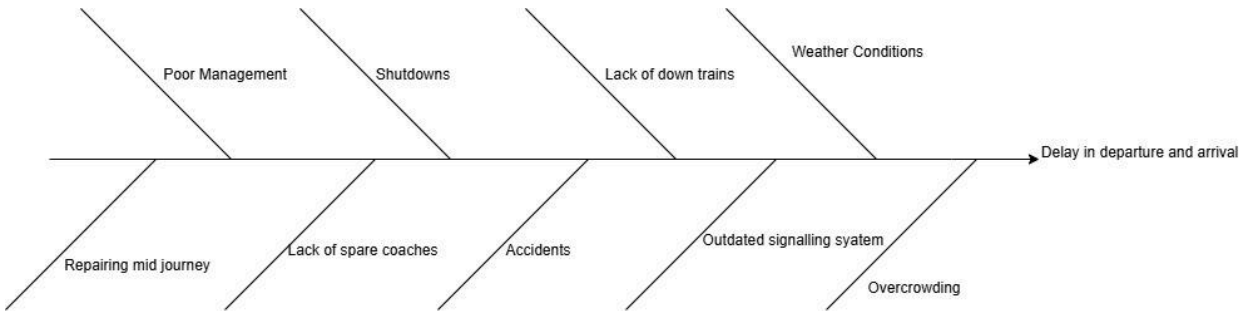
## Step# 2: Root Cause Analysis

**Problem:** Delay in the departure and arrival of the passenger trains.

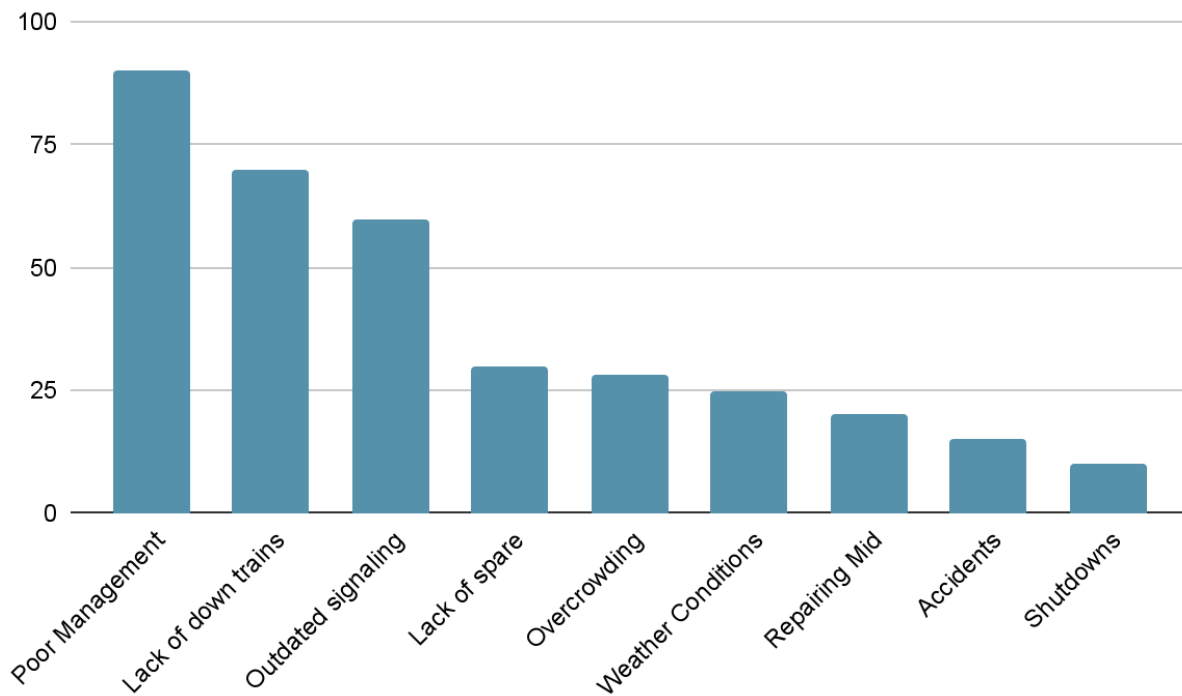
### **Possible Root Causes:-**

- Poor Management (lack of coordination between different departments).
- Overcrowding
- Lack of down trains (trains that return to the location from which up trains came).
- Outdated Signaling System. (The signaling system is used to communicate live with the trains on the lane).
- Lack of spare coaches in case of emergency.
- Bad weather conditions.
- Any accidents that have occurred on the track.
- Any shutdowns that are imposed by the government in some areas.
- Repairing of trains or tracks mid-journey.

**Fishbone diagram:-**



**Pareto chart:-**



### **Poor Management:-**

The reason why poor management is the one with the most weightage is that every function of railways depends on the management. There are several departments that are encapsulated by the management. The list of the departments is as follows:-

- Operations Department
- Maintenance Department
- Traffic Control Department
- Signaling and Communication Department
- Resource Management Department
- Customer Service Department
- Security and Safety Department
- Freight Management Department
- Finance Department

All of these departments are interrelated with each other and cannot work properly if there is a delay in any kind of communication between them. Poor management basically means poor coordination between departments. And when there is poor coordination there will be poor communication, which will lead to the loss of quality information which was required on the specific time and has very low value when the time passes by. So when the management lacks coordination it will lead to dysfunction of various departments and eventually it will lead to delays in the arrival and departure of trains.

#### Output of root cause analysis:-

| <b><u>Element</u></b>            | <b><u>Description</u></b>                                                                                                                                                                                                                                         |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>The problem of...</b>         | Poor Management                                                                                                                                                                                                                                                   |
| <b>Affects...</b>                | <ul style="list-style-type: none"> <li>• Travelers</li> <li>• Railway Workers</li> <li>• Ministry of Finance</li> <li>• Railway Management</li> </ul>                                                                                                             |
| <b>And results in...</b>         | <ul style="list-style-type: none"> <li>• Dysfunction of different departments.</li> <li>• Delay in the arrival and departure of trains.</li> </ul>                                                                                                                |
| <b>Benefits of a solution...</b> | <ul style="list-style-type: none"> <li>• Proper coordination between departments.</li> <li>• Proper functioning of departments.</li> <li>• Satisfaction of customers.</li> <li>• Healthy generation of revenue.</li> <li>• No delays in train timings.</li> </ul> |

## Step# 3: Stakeholder identification

**Questions To Answer:** In order to find out who the direct users of our system are and the other stakeholders, we need answers to a set of questions.

- Who are the users of the system?
  - a. Public passengers
  - b. Railway staff (Drivers, maintenance workers)

- c. Train food and retail vendors
- d. Railway Office workers
- Who is the customer (economic buyer) for the system?
  1. Ministry of Railways of Pakistan
- Who else will be affected by the outputs the system produces?
  1. Station workers (Vendors & maintenance workers)
  2. Ministry of Finance (Economy wise)
- Who will evaluate and approve the system when it is deployed?
  1. The Directorate of Public Relations
  2. Ministry of Railways of Pakistan
- Are there any other internal or external users of the system whose needs must be addressed?

No
- Who will maintain the new system?
  1. Ministry of Railways of Pakistan
- Is there anyone else who cares?

No.

**Users & Stakeholders Table:-**

| Users                     | Other Stakeholders                             |
|---------------------------|------------------------------------------------|
| Public passengers         | Ministry of Railways - Pakistan                |
| Train drivers             | Station workers(vendors & maintenance workers) |
| Train maintenance workers | Ministry of Finance - Pakistan                 |
| Food & retail vendors     | The Directorate of Public Relations - Pakistan |
| Railway office workers    |                                                |

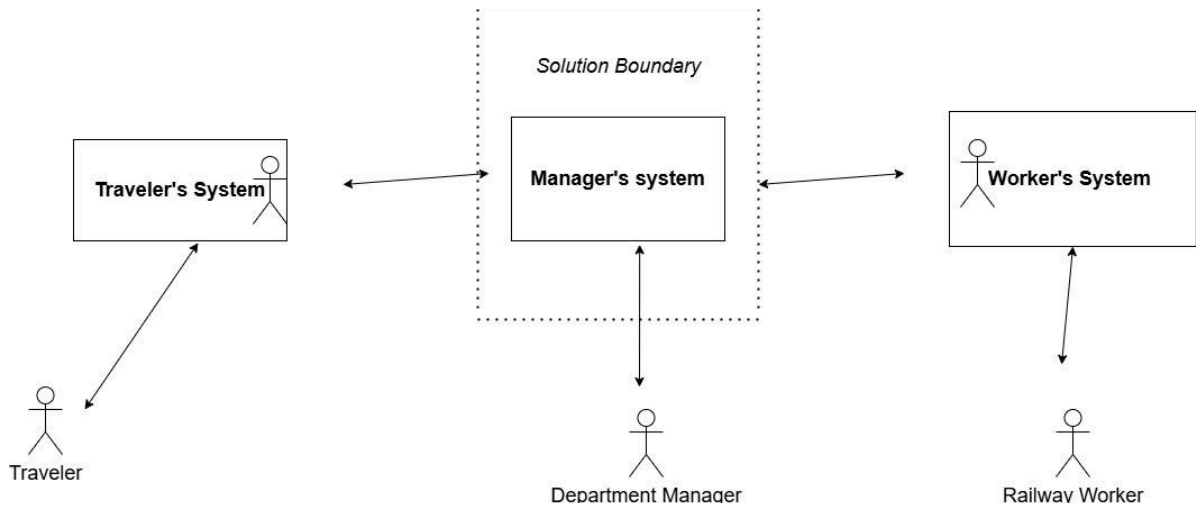
## Step# 4: Solution System Boundary

The solution of this system is bound to the functionality provided only to the managers of the railways. So our solution domain will fall on the manager's module in the system.

### Follow-up questions:-

- Who will supply, use, or remove information from the System?  
Managers.
- Who will operate the system?  
Managers
- Who will perform any system maintenance?  
Software developers.
- Where will the system be used?  
Pakistan Railways
- Where does the system get its information?  
Workers of the interrelated departments.
- What other external systems will interact with the system?  
Traveler's system, worker's system.

### Diagram:-



- **Traveler's System:** Used for booking, Checking train routes, etc.
- **Manager's system:** Manages various departments and communicates information between them
- **Worker's System:** Stores the information of tasks done by workers and sends it to the Manager's System.

## Step# 5: Constraints Identification

| Source     | Constraint                                                                       | Rationale                                                                                                          |
|------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Technology | Implementation of different signaling technologies for usage in different areas. | There does not exist harmony in the type of technology used for signaling of trains in each area.                  |
| Technology | Not all areas will be able to be covered by the software implementation.         | Some areas covered by Pakistan Railways still use semi-manual means of signaling, like the neal ball token method. |
| System     | The software should be compatible with legacy                                    | Pakistan Railways uses outdated systems.                                                                           |

|             |                                                                                            |                                                                                                                                              |
|-------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
|             | systems.                                                                                   |                                                                                                                                              |
| System      | The software should maintain compatibility with existing digital applications.             | There already exists a digital solution to track train locations for Pakistan Railways, it goes by the name of “Pak Rail Live”.              |
| Economics   | The development should be done on existing host systems, no new equipment is to be bought. | The team has a limited budget to work with.                                                                                                  |
| Politics    | System and technology constraints applied to the software implementation.                  | The associated political body “Ministry of Railways” has not approved modernized systems for use. Thus, legacy systems are used to this day. |
| Environment | Necessary implementation of Urdu language as well as English language in software.         | Urdu and English are legally used official languages in Pakistan.                                                                            |

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# Originality report

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COURSE NAME

SE202T- Software Requirement Engineering

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FILE NAME

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## Summary

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1 of 2 passages

Student passage    FLAGGED

- **Who else will be affected by the outputs the system produces?**

### Top web match

**Who else will be affected by the outputs that the system produces?** Who will evaluate and bless the system when it is delivered and deployed? Are there any ...

Task Descriptor: Develop Vision [https://home.iscte-iul.pt/~hro/RUPSmallProjects/core.base\\_rup/capabilitypatterns/develop\\_vision\\_62E594E5.html](https://home.iscte-iul.pt/~hro/RUPSmallProjects/core.base_rup/capabilitypatterns/develop_vision_62E594E5.html)

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2 of 2 passages

Student passage    FLAGGED

- **Are there any other internal or external users of the system whose needs**

### Top web match

Who will evaluate and bless the system when it is delivered and deployed? **Are there any other internal or external users of the system whose needs** must be addressed? Who will maintain the new system?

Task Descriptor: Develop Vision [https://home.iscte-iul.pt/~hro/RUPSmallProjects/core.base\\_rup/capabilitypatterns/develop\\_vision\\_62E594E5.html](https://home.iscte-iul.pt/~hro/RUPSmallProjects/core.base_rup/capabilitypatterns/develop_vision_62E594E5.html)

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